Computability & Logic (prerequisites)
Exercises on Sets and Induction

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From John C. Martin book: read sections 1.1 and 1.2. Also read Examples 1.15, 1.23 and 1.24

Advice

There are more exercises here than you can reasonably complete in one exercise class. Accordingly, we recommend starting with the following exercises:

1.7, 1.8, 1.14, 1.45, and the final exercise ("Proving with Quantifiers").

Once you have completed these, use the remaining exercises to strengthen your understanding of topics you struggled with. For instance, if you find 1.14 difficult, you should try 1.10 and 1.15 for extra practice.

Theoretical Exercises

Sets:

- 1.7: the goal is to use the so-called "set comprehension" notation for sets of sets of numbers.
- 1.8: the goal is to use the set operations (union $\cup$, intersection $\cap$, difference $-$, complement $'$).
- 1.9(e-h): the goal is to understand infinite set unions and set intersections. Prove your answers. An equality ($=$) between sets is proved using 2x subset ($\subseteq$ and $\supseteq$).

Cartesian Products ($A \times B$) and Powersets (2^A):

- 1.14: the goal is to prove equality between sets. Make sure you understand the definition of $A \times B$.
- 1.10: (EXTRA) the goal is to understand powerset notation. Actually, the brackets are as in $2^{(2^{\{0,1\}})}$.
- 1.12a: (EXTRA, if time left) the goal is to get familiar with sets of sets.
- 1.15: the goal is to do sound reasoning about powerset, union, intersection, complement.

Induction:

- 1.45: the goal is to exercise with standard induction over natural numbers.
- 1.47 (EXTRA exercises with induction, only if time left)
- 1.48 (EXTRA exercises with induction, only if time left)
- 1.58: the goal is to exercise with strong induction over natural numbers.

Proving with Quantifiers: In the following proof, the proof structure should be explicit on how you reason about the universal ($\forall$) and existential ($\exists$) quantifiers. You may use common facts about arithmetic.

- Prove (in $\mathbb{N}$): $\forall a (\exists n{\geq}0\, (n^3 + 3n^2 + 3n = a) \iff \exists m{>}0\, (m^3 - 1 = a))$

Phrased using English words rather than mathematical symbols: show that for all $a \in \mathbb{N}$ that if there exists some $n \geq 0$ such that $n^3 + 3n^2 + 3n = a$, then there exists some $m \geq 0$ such that $m^3 - 1 = a$. (Do you see how to translate back and forth between these expressions?)